

Tactile-Enhanced Runtime Learning for Robust Quadruped Locomotion in Wild Environments

Jasmeh Singh and Changyong Cao

Abstract

This paper presents a novel cyber-physical system (CPS) for enhancing quadruped robot locomotion in dynamic wild environments by integrating a custom tactile foot sensor with a hybrid machine learning (ML) framework. Building on the runtime learning architecture of (author?) [1], we address critical limitations in foot sensing, physics modeling, terrain characterization, and safety learning by developing a piezoresistive tactile sensor for the Unitree Go2 robot and extending the high-performance (HP)- Student/high-assurance (HA)-Teacher framework. The sensor captures ground reaction forces and friction coefficients, enabling real-time terrain classification with less than 10% error. The framework incorporates tactile data into sensing, dynamics models, reward functions, navigation, and teaching-to-learn paradigms, achieving a reduction in fall rates and 20-40% improvement in navigation success compared to state-of-the-art baselines. Evaluations in Nvidia Isaac Gym and real-world wild terrains, such as forest trails and rocky slopes, demonstrate enhanced stability, adaptability, and energy efficiency. The system's robustness, validated through tactile-driven performance metrics, advances CPS for safety-critical applications, including search and rescue and environmental monitoring. Open-source contributions, including sensor designs and ML code, are provided to foster further research in autonomous robotics.

Introduction

Quadruped robots, such as the Unitree Go2, are increasingly critical for tasks in unstructured wild environments, such as search and rescue, exploration, and disaster response [2, 3]. However, navigating dynamic terrains with varying properties, such as slippery surfaces, soft soil, or uneven rocks, poses significant challenges due to limitations in existing sensing and control systems. The runtime learning framework by (author?) [1] integrates a deep reinforcement learning (DRL)-based high-performance (HP)-Student with a physics-based high-assurance (HA)-Teacher, enabling adaptive locomotion with safety guarantees. Despite its strengths, the framework suffers from four key limitations: (1) lack of tactile foot sensing, relying solely on proprioceptive (e.g., IMU) and exteroceptive (e.g., depth camera) data; (2) static physics model assumptions that fail to capture complex terrain dynamics; (3) oversimplified terrain characterization using height-based occupancy maps; and (4) a teaching horizon that does not account for terrain-specific safety constraints. To address these limitations, we propose a novel CPS that enhances quadruped locomotion by integrating a custom tactile foot sensor with a hybrid ML framework. The sensor, designed for the Go2's footpads, captures ground reaction

forces ($\mathbf{f}_i \in \mathbb{R}^3$) and friction coefficients (μ_i), enabling real-time terrain classification. The framework extends the HPStudent/HA-Teacher architecture by incorporating tactile data into: (1) the sensing module via an augmented state vector $\mathbf{s}_{\text{foot}}$, (2) dynamics models with terrain-specific contact dynamics, (3) safety sets with force and friction constraints, (4) reward functions with terrain-adaptive terms, (5) navigation with tactile-enhanced cost maps, and (6) teaching-to-learn with terrain-adaptive horizons. Evaluations in simulation (Nvidia Isaac Gym) and realworld wild environments demonstrate a 50% reduction in fall rates and 30% improvement in navigation success compared to baselines, alongside enhanced energy efficiency. Contributions present in this paper:

- A custom piezoresistive tactile foot sensor for the Unitree Go2, achieving less than 5% force error and 10% friction error, optimized for wild environments.
- A hybrid ML framework integrating tactile data into sensing, dynamics, navigation, and teaching modules for robust locomotion.
- A terrain-adaptive teaching-to-learn paradigm with dynamic horizons, enhancing safety policy learning.
- Comprehensive validation in simulated and real-world settings, with open-source sensor designs and code to advance CPS research.

The paper is organized as follows: Section 2 reviews related work; Section 3 details the proposed methods; Section 4 presents experimental setups and results; Section 5 discusses findings and limitations; Section 6 concludes with future directions.

2 Related Work

Recent advances in quadruped locomotion leverage DRL for adaptive policies [2] and safetycritical control for robust navigation [3]. Tactile sensing, while prevalent in manipulation [4], is underexplored in quadruped robots, with most systems relying on proprioceptive (e.g., IMU) and exteroceptive (e.g., depth cameras) data. The runtime learning framework by (author?) [1] combines a DRL-based HP-Student with a physics-based HA-Teacher, achieving real-time adaptation but lacking tactile foot sensing, dynamic physics models, and terrain-specific navigation. Height-based occupancy maps oversimplify terrain properties, and teaching horizons ignore terrain constraints. Our work bridges these gaps by integrating a custom tactile sensor with a hybrid ML framework, enhancing stability and adaptability in wild environments.

3.1 Custom Tactile Foot Sensor

We designed a piezoresistive tactile sensor array for the Go2’s footpads to capture ground reaction forces ($\mathbf{f}_i \in \mathbb{R}^3$) and friction coefficients (μ_i) for each foot ($i = 1,2,3,4$). The sensor is

optimized for wild environments, with IP67 durability against dust, moisture, and impact, and a force range of 0–500 N to accommodate the Go2’s dynamic loads. Design and Fabrication: The sensor array consists of piezoresistive elements embedded in flexible substrates, designed using CAD (SolidWorks) to match the Go2’s footpad geometry. Finite element analysis ensured uniform load distribution. Prototypes were fabricated via 3D printing and soft lithography, targeting a response time of <10 ms for real-time operation. Calibration: The sensor was calibrated using a force plate, mapping voltage outputs to forces and estimating friction via force ratios ($\mu_i = |f_{i,tangential}|/|f_{i,normal}|$). A random forest classifier was trained on labeled data from lab terrains (rocky, sandy, grassy, wet) to predict terrain types with <10% error, using the tactile state vector: $s_{foot} = [f_1, f_2, f_3, f_4, \mu_1, \mu_2, \mu_3, \mu_4]^T$. Validation: The sensor was tested in lab terrains with dynamic perturbations (e.g., shaking platforms, wet surfaces), achieving force errors <5% and friction errors <10% compared to ground truth. Durability was verified through water immersion and drop tests.

3.2 Tactile Enhance ML Framework

The ML framework extends the HP-Student/HA-Teacher architecture of [1], integrating tactile sensing into the core modules of sensing, dynamics, safety, reward, navigation, and teaching to enable terrain-adaptive locomotion. This tactile integration allows the quadruped to infer surface properties such as stiffness, friction, slope, and roughness from ground contact forces, enabling autonomous gait adaptation similar to how humans instinctively adjust stride on ice, gravel, or slopes.

Sensing Module: We augment the proprioceptive state vector $s = [h, \Theta, v, \omega]^T$ (height, orientation, velocity, angular velocity) with a tactile vector s_{foot} encoding per foot’s ground reaction forces. $F_{grf}(t) = [F_x, F_y, F_z]$, terrain roughness.

$$R_{terrain}(t) = 3.0 \times \| p_{foot}(t) - p_{foot}(t-1) \| / 2$$

and slope,

$$S_{terrain}(t) = 0.5 \times |h_{foot}(t) - h_{base}(t)|$$

The HP-Student’s observation space (Eq. 19) expands from 48 to 60 dimensions:

$$o_t = [z_t, e_t, s_{foot}]^T$$

where $z_{t:t}$ contains terrain classifier outputs and $e_{t:t}$ captures force tracking errors. A Kalman filter reduces sensor noise variance by ~4.8%. Terrain classification uses a rule-based

model: flat ($R < 0.02$, $S < 0.1$), rough ($0.05 \leq R < 0.1$), steep ($S \geq 0.4$), slippery ($\mu < 0.3$), and obstacle ($R \geq 0.15$). Temporal smoothing is applied over a 5-step window:

$$T_{\text{smooth}}(i, t) = \arg \max_k \sum_{\tau=0}^4 I(T_{\text{history}}(i, t - \tau) = k)$$

Dynamics Model: The HA-Teacher's dynamics model (Eq. 9) incorporates tactile-dependent contact dynamics:

$$b_e(t + 1) = A(s(k), s_{\text{foot}}(k)) \cdot b_e(t) + B(s(k), s_{\text{foot}}(k)) \cdot a_{\text{HA}}(t) + h(b_e(t), s_{\text{foot}}(k))$$

with AAA and BBB updated using a Coulomb friction model parameterized by μ_i . The patch center (Eq. 8) is redefined as:

$$e_k^* = \chi \cdot e(k) + e_{\text{foot}}(k)$$

where $e_{\text{foot}}(k)$ provides a terrain-dependent offset (e.g., increased safety margins on slippery surfaces). Gait parameters are dynamically scaled based on terrain type:

$$v_{\text{cmd}} = \alpha_T v_{\text{nominal}}, \quad L_{\text{step}} = \beta_T L_{\text{nominal}}, \quad T_{\text{stance}} = \gamma_T T_{\text{nominal}}$$

Where α_T , β_T , γ_T are terrain-conditioned scaling coefficients. For example, $\alpha_T = 0.6$ reduces commanded speed on slippery terrain, $\beta_T = 1.3$ increases step height on rough ground, and $\gamma_T = 1.2$ extends stance duration on slopes. Control gains are computed using CVX [6].

Safety Constraints: The safety set (Eq. 1) incorporates tactile feedback,

$$\mathcal{S} = \{e \in \mathbb{R}^m \mid -c \leq C \cdot e \leq c, \ |f_i| \leq f_{\text{max}}(T), \ \mu_i \geq \mu_{\text{min}}\}.$$

where f_{max} varies by terrain — 200 N on soft or slippery ground to prevent sinking or slip, and up to 500 N on rigid surfaces. The minimum friction threshold remains $\mu_{\text{min}} = 0.3$, ensuring stable locomotion.

Reward Function: The HP-Student's reward function is extended,

$$R_{\text{adapt}} = -\lambda_1 |v_{\text{cmd}} - v_{\text{meas}}| - \lambda_2 \|F_{\text{grf}} - F_{\text{ref}}(T)\|_2 - \lambda_3 \Delta_{\text{slip}}$$

Here, F_{ref} is a terrain-specific reference force profile, and Δ_{slip} penalizes foot slippage during stance. These terms incentivize dynamic contact stability, terrain-specific adaptation, and energy-efficient movement.

Navigation Module: The cost map (Fig. 3 in [1]) is augmented with tactile data, assigning higher costs to regions with $\mu_i < 0.4$ or significant force variability. The Fast Marching Method (FMM) prioritizes stable, low-slip, and energy-efficient paths, dynamically updating waypoints based on real-time s_{foot} .

Experience tuples (Eq. 19) now include tactile feedback:

$$\mathcal{E}_\tau = \{o(t), a_{\text{HA}}(t), o(t+1), R(o(t), a_{\text{HA}}(t)), s_{\text{foot}}(t)\}_{t=k}^{k+\tau}$$

The teaching horizon τ (Eq. 18) is dynamically adjusted based on terrain complexity (e.g., $\tau = 10$ when $\mu_i < 0.4$), improving terrain-specific safety learning and generalization.

Soft Actor-Critic (SAC): A SAC algorithm [5] is implemented with a terrain-aware policy network taking o_t as input and a critic network estimating Q-values. Model pruning reduces inference time to under 20 ms per step, maintaining real-time control performance.

$\theta=0.1$, $c_\mu=0.5$. The HA-Teacher intervened according to the updated safety set SSS with tactile constraints.

Baselines: We compared the proposed framework with methods from Cai et al. [1]:

- CLF-DRL: Control Lyapunov function-based DRL
- Phy-DRL: Physics-informed DRL
- HA-Teacher: Physics-based safety policy

Metrics: Evaluation used fall rate (falls per 100 m), navigation success rate (successful waypoint reaches), travel time, energy consumption (motor power integral), safety violations (constraint breaches), and terrain adaptation events (successful classifications and gait adjustments).

Results:

- Fall Rate: 0 falls per 100 m vs. 0.5–3 falls per 100 m for baselines (100% reduction).
- Navigation Success: 100% episode success (4/4) with 90.4% goal progress vs. 60–85% for baselines (20–40% improvement).
- Safety Performance: 0 safety warnings and 0 control errors vs. 10–50 warnings per episode for baselines (100% improvement).
- Terrain Adaptation: 5,940 successful real-time terrain classifications and gait adaptations vs. static gait parameters in all baselines.
- Overall System Health: 80.1% system health rating with a “GOOD” performance classification.